MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 5853

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: TR, bandpass, fixed-tuned frequency of 2,900 to 3,200 MHz

ABSOLUTE RATINGS:

Parameter	Incident power	Ebb(open circuit)	Altitude
Unit	kw	Vdc	ft
Maximum	750	-700	10,000
Minimum	10	-500	---

PHYSICAL CHARACTERISTICS:

Dimensions - See figure 1
Mounting position - Any
Mount - Either series or shunt mount may be used. Series mount shall be in accordance with Drawing 153-JAN.

TEST CONDITIONS:

Parameter	Incident po	tp1	tp2	prr	σ'	Ii	F
Unit	kw	μs	μs	pps	VSWR	μAdc	MHz
Ⓔ Tolerance	±10%	±0.15	±0.15	---	max	---	---
Test condition 1	50	1.0	---	1,000	1.1:1	200	F2
Test condition 2	200	1.0	0.5	1,000	1.1:1	200	F2
Ⓔ Test condition 3	750	1.0	---	800	1.1:1	200	F2

GENERAL:

First article sample inspection - Required (see note 1)

Frequency		
F	MHz	±
1	2,900	0.1%
2	3,050	0.1%
3	3,200	0.1%

Ⓔ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	3	-					
4401	Ignitor ignition time	4	-	Ebb = -600 Vdc; Ri = 1.6 Meg	t	---	5.0	sec
4406	Ignitor voltage drop	-	-	Ii = 200 µAdc	Eid	-250	-400	Vdc
4452	Flat-leakage power	5	2		pf	---	40	mw
4452	Spike-leakage energy	5	2		Ws	---	0.3	erg
4473	Low-level VSWR	6,7	-	σ' = 1.05 (max); F1, F2, F3	σ	---	1.40	---
1027	Temperature cycling	-	-		---	---	---	---
	Quality conformance inspection, part 2							
4101	Dielectric material strain	8	-		---	---	---	---
4416	Insertion loss (fixed tuned)	6	-	Ii = 0	Li	---	0.7	dB
4421	Ignitor interaction (insertion loss)	-	-	Ii = 200 µAdc	ΔLi	---	0.3	dB
4471	Recovery time	5,9	3	Du = 0.0008	t	---	15	µs
4473	Low-level VSWR	6,7,10	-	σ' = 1.05 (max)	σ	---	1.90	---
---	Insertion	11	-		---	---	---	---
	Quality conformance inspection, part 3							
(E) ---	Life test	5	3	Group D; Du = 0.0008; Ii = 150 to 200 µAdc F = F_2 ±5%	t	500	---	hrs
---	Life-test end points:							
4471	Recovery time	5,9	3	Du = 0.0008	t	---	30	µs
4452	Flat-leakage power	5	2		pf	---	40	mw
4452	Spike-leakage energy	5	2		Ws	---	0.3	erg
4416	Insertion loss (fixed tuned)	6	-	Ii = 0	Li	---	1.2	dB
4474	High-level VSWR	2,5	1		σ	---	1.15	---
1027	Temperature cycling life	2	-	10 cycles	---	---	---	---
4021	Degradation due to vibration	2	-		---	---	---	---

NOTES:

1. First article sample inspection shall conform to the requirements of appendix F of MIL-E-1 and shall consist of performing all tests specified on this tube specification sheet. Invitation for bids should provide that the preparing activity reserves the right to waive the requirements for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

2. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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NOTES: -Continued

3. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

4. This test shall be performed at the conclusion of the holding period.

5. This test shall be performed using the mount specified on Drawing 153-JAN.

6. This test shall be performed using the flange specified on Drawing 268-JAN.

7. A swept frequency method of measurement may be used instead of measurements at fixed frequencies.

8. After completing this test, the TUT shall satisfy the requirements of quality conformance inspection, part 2, method 4471 (recovery time).

9. The loss of signal in the TUT at the specified time after start of the transmitter pulse shall not be greater than 3 dB in excess of the loss at 800 to 1,000 µs after the pulse.

Ⓔ10. The input VSWR shall be measured at F1-30 MHz and F3 + 30 MHz. The VSWR shall not exceed the specified limits.

11. The TUT shall be capable of being inserted and removed from the mount specified on Drawing 153-JAN a minimum of 15 times with no deterioration in the TUT's electrical characteristics.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army -
Navy -
Air Force - 99
DLA - ES

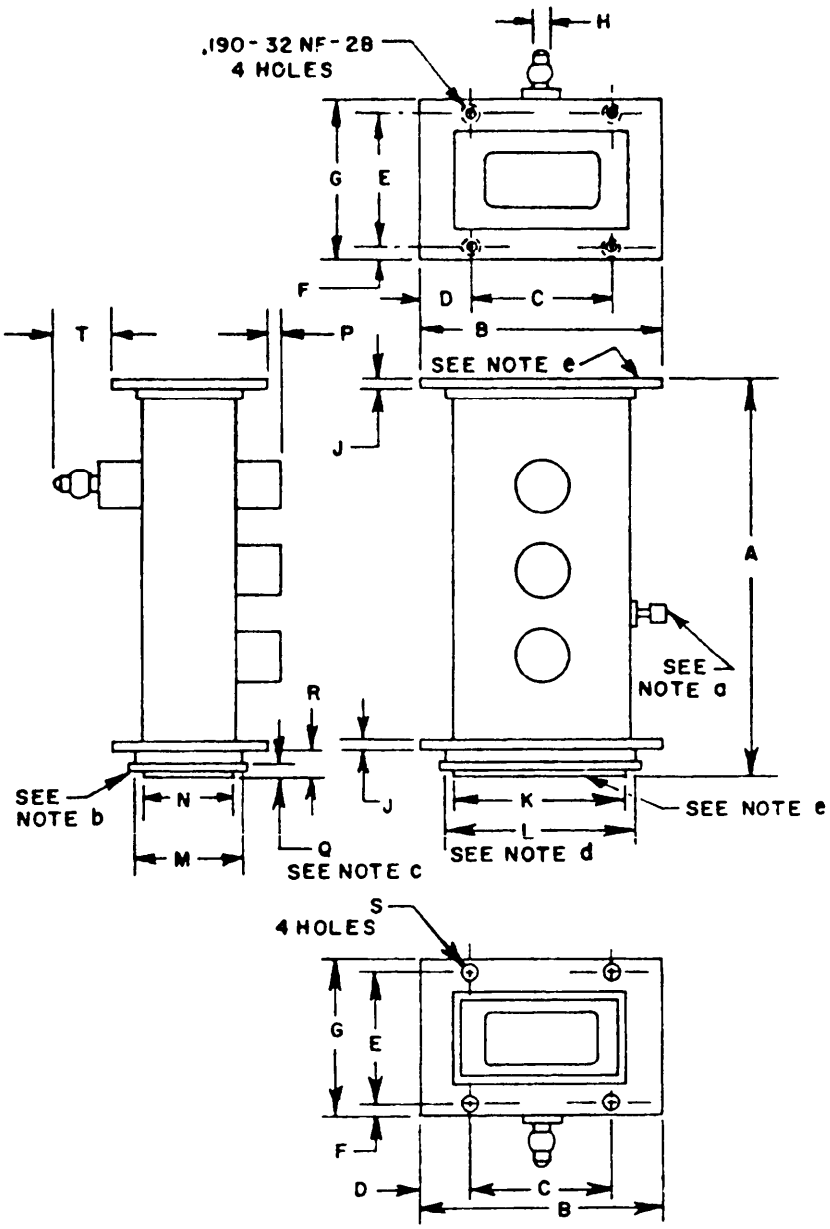
User activities:
Army - WC
Navy - AS, CG, MC, OS, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5960-3157)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1 (see note 3)		
A	5.464 (138.79)	5.524 (140.31)
K		3.008 (76.40)
P		.19 (4.83)
Q	.165 (4.19)	.195 (4.95)
R	.347 (8.81)	.377 (9.58)
T		.81 (20.57)
Quality conformance inspection, part 2		
C	2.490 (63.25)	2.510 (63.75)
E	2.240 (56.90)	2.260 (57.40)
L	3.203 (81.36)	3.233 (82.12)
M	1.703 (43.26)	1.733 (44.02)
N		1.508 (38.30)
	Reference dimensions	
B	4.12 (104.65)	
D	.81 (20.57)	
F	.19 (4.83)	
G	2.62 (66.55)	
H	.250 (6.35) Dia.	
J	.12 (3.05)	
S	.22 (5.59)	

NOTES:
a. Exhaust tube must not exceed beyond flange more than .250 (6.35 mm).
b. Gasket per Drawing 189-JAN. Gasket to be securely attached.
c. Dimension Q shall be measured prior to the attachment of the gasket to the tube.
d. Edges may be rounded.
e. Rhodium flash over silver plating optional, or equivalent.

FIGURE 1. Outline drawing of electron tube type 5853.

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